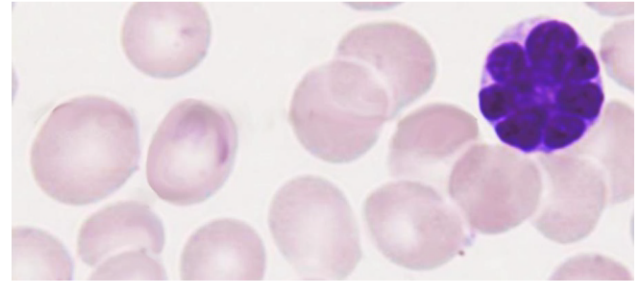


- ↑ N:C
- Indentation in nucleus gives appearance of flower - **Flower cells**
- Adult T cell Leukemia caused due to association with **HTLV-1**

ADULT T-CELL LEUKEMIA



ETIOLOGY

00:05:52

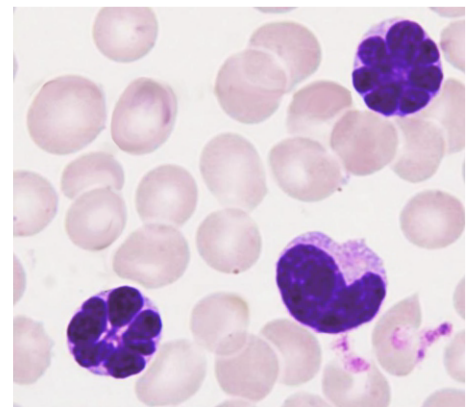
A	• Ataxia Telangiectasia
B	• Bloom Syndrome (defective DNA repair)
C	• Chemotherapy induced secondary malignancy • Drugs responsible - alkylating agents like Cyclophosphamide, Topoisomerase II inhibitor ◦ Drugs used in T/t of AML, Myelodysplastic Syndrome, Carcinoma breast
D	• Down syndrome
I	• Ionizing radiation, Infection- HTLV
F	• Fanconi anemia
K	• Klinefelter syndrome
N	• Neurofibromatosis type 1

- Philadelphia chromosome, t(9:22)
- Other causes
 - t(8:14) → Burkitt's Lymphoma
 - t(4:11)
 - t(1:19)

HTLV-1

00:10:50

- Adult T cell Leukemia
- Flower cell or clover leaf



- Increased ICP/Meningismus
- Cell Count $>5 \text{ cells/mm}^3$
 - Lumbar puncture → breach of blood vessels → blood can enter CSF → cancer cells introduced → intrathecal methotrexate given after lumbar puncture to minimize chances of accidental inoculation of cancer cells into CSF
- CT - Leptomeningeal metastasis
- Intrathecal methotrexate should be given after LP

CLINICAL PRESENTATION

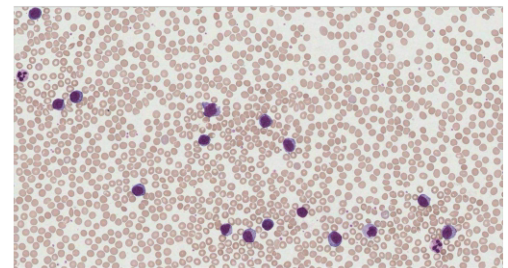
00:13:25

- Age group of presentation - 2-8 years, > 50 years
- Progressive Pallor and Anemia
- Bleeding - Epistaxis
- Infection - Recurrent Pneumonia
- Petechia, purpura, bleeding
- Hepatosplenomegaly, Lymphadenopathy
- CNS leukemia → $\uparrow$ ICP → 6th nerve palsy
- Bony/Sternal Tenderness

WORK UP

00:18:00

- CBC and Peripheral smear
 - Hb $\downarrow$
 - TLC Normal/ $\downarrow$ / $\uparrow\uparrow$
 - Leukostasis (AML $>$ ALL) → initially low neutrophil count → recurrent pneumonia
 - Platelet count $\downarrow$
- IOC - Bone Marrow Aspiration/Biopsy



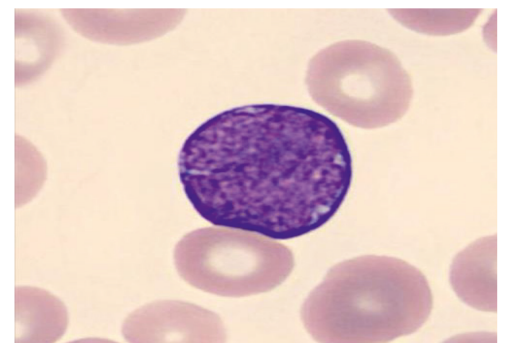
Bone marrow smear: ALL

BONE MARROW ASPIRATION/BIOPSY

Salah and Klima
needles used

Site-Posterior Superior
Iliac Spine

- Lymphoblasts count $>20\%$ with $\uparrow$ N:C ratio, fine chromatin



TEST DONE ON BONE MARROW SAMPLE

CELL MORPHOLOGY EVALUATED IN SMEARS USING WRIGHT AND GIEMSA STAIN

- L1 (M/c)
- L2
- L3 -
 - Burkitt's leukemia and associated with t(8:14)
 - Mature B cell ALL
 - CD10 $\oplus$ with S. Ig $\oplus$



IMMUNOPHENOTYPING USING FLOW CYTOMETRY

- B cell ALL: CD19 CD22 CD79A CD10
- T cell ALL: CD2 CD3 CD7
- Biphenotypic variety: combining both B cell ALL and T cell ALL

CYTOGENETICS: FISH (FLUORESCENT IN SITU HYBRIDISATION)

- t (9:22)
- t (8:14)
- t (4:11)
- t (1:19)

CYTOCHEMISTRY (DIFFERENTIATE MYELOBLAST FROM LYMPHOBLAST)

- MPO- **negative**
- PAS- Positive
- Terminal deoxynucleotidyl transferase (TdT)- Positive

- Serum electrolytes and uric acid to get baseline values due to risk of Tumor Lysis syndrome
 - **Tumor lysis syndrome** - chemotherapy → death of large number of cancer cells → ATP breakdown to produce phosphates → hyperphosphatemia → chelation with cancer → calcium ↓
 - Potassium ↑
 - Uric acid ↑ → blockage of kidney tubules by uric acid crystals → acute tubular necrosis

T/T OF TUMOR LYSIS SYNDROME

Allopurinol-inhibit uric acid production by inhibiting xanthine oxidase

I/V Fluid for hydration

Rasburicase-DOC

- Lumbar puncture - to check CSF for leukemia cells
- HLA testing as patient subsequently sent for stem cell transplantation

HIGH-RISK ALL FEATURES

00:26:20

- Age group: <1 year and >10 years (extreme age)
- ↑ WBC count: >50,000/mm³
- Lymphadenopathy/ Hepatosplenomegaly/ Mediastinal mass (organomegaly)
- Mature-B Cells
- **Hypoploidy**
- t(9:22), t(8:14), t(4:11), t(1:19)
- Blasts: >1,000/cumm in peripheral smear after 14 days of chemotherapy
- Absence of CD₁₀ and presence of MLL rearrangement
- T/t - Allogenic stem cell transplantation in 1st remission with chemotherapy

LOW RISK ALL

- Age group: 1-9 years
- WBC count <50,000/mm³
- Pre-B cell ALL
- **Hyperploidy**

STANDARD RISK

- Same as high risk +normal cytogenetics

VERY HIGH RISK

- Induction failure
- t(9:22)

Complete Molecular Remission

- $<0.01\%$ or <1 leukemia blasts/10,000 for normal cells
- Complete haematological resolution
 - $<5\%$ Blast in bone marrow
- Patient with ALL, Philadelphia Chromosome $\oplus$
 - Tyrosine kinase inhibitors: **2nd generation** used as they cross blood brain barrier and $\uparrow$ risk of development of CNS Leukemia $\rightarrow$ Dasatinib, Nilotinib used
- **CAR-T** cell therapy targeting CD 19 (Chimeric antigen receptor target): for relapsed/refractory CD 19 ALL using Tisagenlecleucel and Blinatumomab

CHEMOTHERAPY FOR ALL

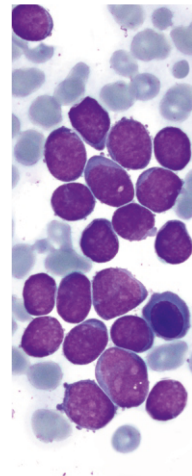
- **Pre phase** -
Prednisone/
Dexamethasone plus
Vincristine/Cyclophosphamide
◦ To $\downarrow$ risk of tumor lysis syndrome
- **Induction** - (hyper CVAD)
Hyperfractionated
Cyclophosphamide +
Vincristine + doxorubicin
and dexamethasone
- **Post remission consolidation** - $\uparrow$
dose methotrexate
and Cytarabine
- **Maintenance** -
6MP +
Methotrexate +
intrathecal
Methotrexate
◦ For 2-2.5 years



Q. A 6-year-old child presents with fatigue, pallor, and nasal bleeds for the past few weeks. Physical examination reveals hepatosplenomegaly. Laboratory investigations show NCNC anemia with granulocytopenia and low platelet count. Bone marrow aspiration shows hypercellularity with more than 20% blasts image of which is shown. Which of the flow cytometry markers will be seen in this patient?

- a. Cd10
- b. Cd16
- c. Cd33
- d. Cd38

Ans. (a)





BRAID MULTIPLE MYELOMA



MULTIPLE MYELOMA (MM)

00:01:28

- Plasma cell dyscrasia
- M/c cause in Multiple myeloma: **Infection** > renal failure
- M/c abnormal Immunoglobulin produced in MM: **IgG (kappa)** → 60%
- Best imaging modality for bone marrow involvement in MM: Whole body DW-MRI
- Best imaging modality for diseases activity or MRD (minimal residual diseases) in MM: PET/CT SCAN
- Lytic lesions of vertebrae → vertebral body collapse → compress spinal cord → compressive myelopathy



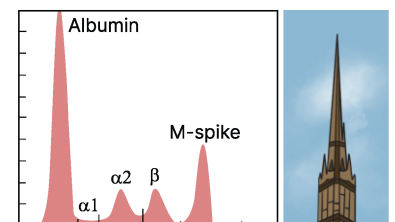
PET/CT SCAN

Lytic lesion in vertebrae

PATHOGENESIS

00:04:11

- Normal heavy chain isotypes: Ig G, A, M, E, D
- Light chain isotypes: kappa, lambda
- Normal light chains assay - <10 mg/day
- Characterized by
 - Defective immunoglobulin production: UTI, Pyelonephritis, Pneumonia
 - ↑ viscosity of blood causing sluggish blood flow (normal viscosity 1.8x of water)
 - Hyper viscosity - value is > 4x of water
 - Hyperviscosity is due to ↑ production of light chain of antibodies
 - Bence jones proteins (Positively charged proteins in urine) → not detected on urine dipstick
 - Only negatively charged proteins identified on dipstick and proteins in multiple myeloma is **positively charged**- Paraproteins
 - Paraproteins documentation done by Urinary Protein Electrophoresis
- Plasma cells produce cytokine that causes Osteoclasts activation
 - Cytokine: Macrophage Inflammatory Protein-1 alpha (MIP-1 alpha)
 - produces Hypercalcemia and bony lytic lesions at - vertebrae > pelvis > ribs > skulls
- Excess of light chains and subtypes detected by 2 techniques

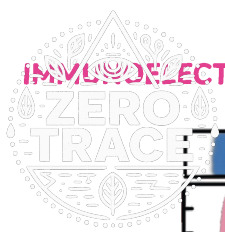


Immunoelectrophoresis

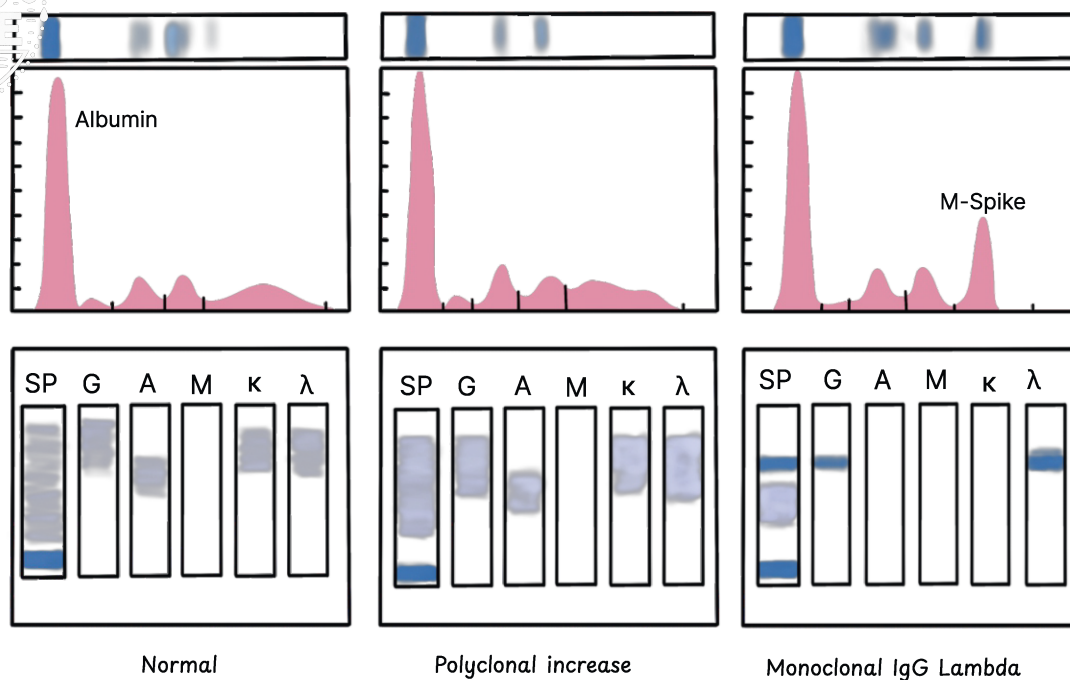
Immunofixation

- Shows **M spike / Church spire** appearance
- M spike - monoclonal antibody, indicates tumor load >10⁹ cells

Shows subtype of light chain overproduced like kappa and lambda



IMMUNOELECTROPHORESIS+ IMMUNOFIXATION



- Polyclonal increase: Ig G, A $\uparrow$
- Monoclonal IgG lambda: $\uparrow$ gamma globulin on serum protein electrophoresis (lambda variety)

HYPOGAMMAGLOBULINEMIA

- Overall antibody production $\downarrow$
- Defective antibody production $\uparrow$

KIDNEY DAMAGE IN MM

- **Hypercalcemia:** leading cause of damage to kidneys in Multiple Myeloma
- Hyperviscosity causes tubular damage
- Bence jones proteins damage tubule lining
- Amyloidosis
- Infections: Pyelonephritis
- Long term intake of NSAIDs for bone pain

ETIOLOGY

00:13:30

- Radiation can lead to development of all blood cancers except for Chronic Lymphocytic Leukemia
- Woodworkers & farmworkers: **Petrochemical exposure**
- Hyperploidy
- Chromosomal translocations: t(13q:14), t(11:14), t(4:14), t(14:16)
- Deletion of 17p
- 1p amplification and 1q deletion

TRIPLE HIT MULTIPLE MYELOMA (Based on Mayo stratification for myeloma on basis of which T/t is made)

- Refers to high risk subtype of multiple myeloma characterized by presence of 3 high risk cytogenetic abnormalities in same clone, such as del(17p), t(4:14), t(14:16) or 1q21 gain

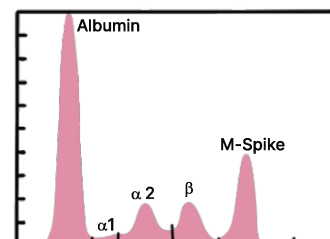
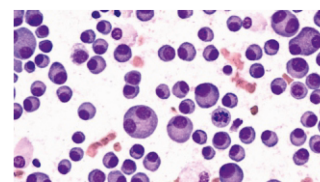
RISK STRATIFICATION IN MULTIPLE MYELOMA

CATEGORY	DEFINITION	COMMON CYTOGENETICS	PROGNOSIS
Standard risk	No high-risk cytogenetics	t(11:14), t(6:14)	Good
High risk	Presence of any one high risk cytogenetic abnormality	Del(17p), t(4:14), t(14:16), 1q21 gain	Poor
Double hit	2 high risk cytogenetic abnormalities in same clone	Del(17p) +(4:14)	Very poor
Triple hit	3 high risk cytogenetic abnormalities in same clone	Del(17p) + t(4:14) + 1q21 gain	Extremely poor

TRIAD OF MULTIPLE MYELOMA

00:16:15

- Bone marrow biopsy: **IOC**
 - Absolute plasma cell in BM: >10%
- Screening test: Serum Protein Electrophoresis, Urinary Protein Electrophoresis: Shows presence of M-Spike (Monoclonal spike)
- Features: (any 1 present)
 - C** - Hypercalcemia (calcium >11 mg)
 - R** - Renal failure
 - Creatinine clearance
 - Calculation of Estimated GFR
 - Serum creatinine (>2mg)
 - A** - Anemia
 - Normocytic normochromic anaemia (Hb <10gm%)
 - B** - Bony lytic lesions → demonstration by MRI
 - B** - Biomarkers of malignancy



BIOMARKERS OF MALIGNANCY (mnemonic: SLiM)

- Any 1 or more of these

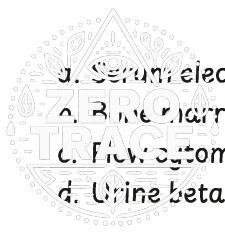
S	Sixty percent of plasma cell
Li	Involved free light chains >100 mg/dl or uninvolved to involved free light chain ratio >100
M	At least 1 focal lesion by MRI of >5 mm

- Multiple myeloma → ≥10% Plasma cells with CRAB/ SLiM, ≥60% even without CRAB

CASE SCENARIO

00:21:01

Q. 70-year-old man presents with lethargy, progressive pallor and low back pain for past 6 months. MRI spine shows compressive myelopathy. He complains of vertigo and dizziness on getting up. Work up shows Hb 8 gm%, normocytic normochromic with normal RBC Count, S. bilirubin is 0.6 mg/dl with low albumin globulin ratio, BUN is 80 mg/dl and serum creatinine is 1.6 mg/dl and serum calcium 11 mg/dl. What is next step in diagnosis of this condition?



- Serum electrophoresis
- Bone marrow biopsy
- Flow cytometry on peripheral blood
- Urine beta 2 microglobulin levels

Ans. a

Explanation (Differential diagnosis- Multiple Myeloma)

- Low back pain in an old man points towards
 - Carcinoma prostate
 - Multiple myeloma
- Vertigo and dizziness – explained by Hyperviscosity
- Normocytic normochromic anaemia

Important Information

- Plasma cell leukemia: ≥ 2000 plasma cells/cumm in blood
- $>20\%$ of plasma cells in peripheral smear out of all leucocytes

WORK UP

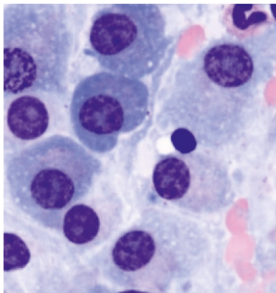
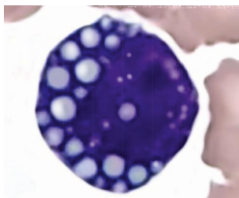
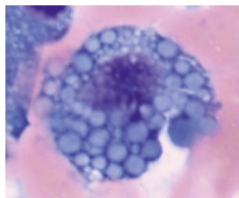
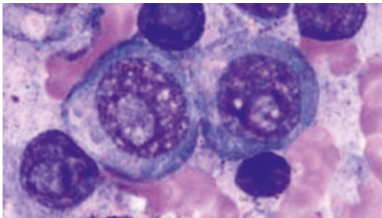
00:24:31

CBC	• Hb ↓ • ESR: ↑ (100 mm fall in 1st hour- also present in Giant cell arteritis)
A:G RATIO	• ↓ and ↓ serum albumin
KFT	• Deranged due to hypercalcemia, sluggish blood flow to kidney tubules, pain killers use
WHOLE BODY DW-MRI	• To demonstrate bone marrow involvement
PET CT	• In case of diseases burden, activity, minimal residual diseases
COAGULOGRAM	• Can be normal or slightly Deranged in case liver gets affected • ↑BT due to thrombocytopenia as megakaryocyte does not get time and space to multiply as plasma cells growing uncontrollably
ANION GAP	• Since paraproteins are positively charged, kidney will try to ↑ negative charges to counterbalance and maintain electroneutrality, causes kidney absorb more chloride ions back in blood • Reduced anion gap
SERUM CALCIUM	• ↑ due to osteoclastic activity
PHOSPHATE	• Normal/ ↓
SERUM ALKALINE PHOSPHATASE	• Normal as MIP 1 alpha contributes directly to damage to osteoclast

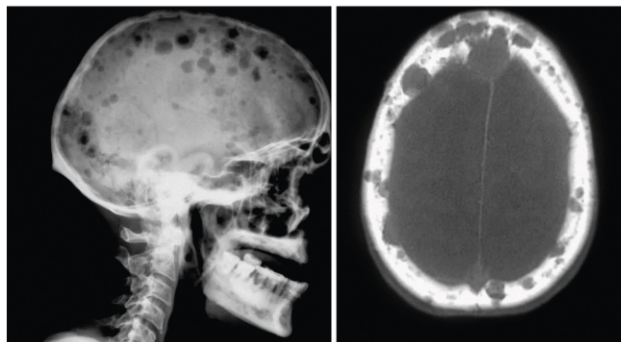


SCREENING TEST FOR MULLEN'S	- Serum protein electrophoresis > urine protein electrophoresis showing M spike - Immunofixation
IOC - BONE MARROW BIOPSY	>10% plasma cells
FLOW CYTOMETRY (BONE MARROW SPECIMEN)	CD138
SERUM LIGHT CHAIN ASSAY	>100 mg/dl
URINE BETA 2 MACROGLOBULIN	For prognosis
AMYLOID FAT PAD BIOPSY	Amyloidosis
FISH	t(13q:14)
X RAY SKULL	Lytic punched out (rain drop) lesion

BONE MARROW BIOPSY FINDING IN MULTIPLE MYELOMA

CLOCK FACE CHROMATIN - Plasma cells with Eccentric nucleus - Clock face chromatin - Chromatin arranged in periphery of cell	
MOTT CELLS Intracytoplasmic inclusions of light chain - Russel bodies Intranuclear inclusions - Dutcher bodies	  Mott cells  Dutcher bodies

- Xray Skull: **Raindrop lesions**

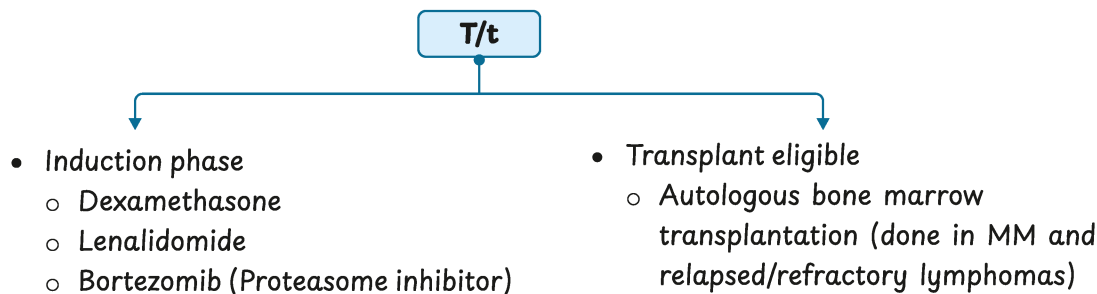


	MGUS	SMOLDERING MYELOMA	MULTIPLE MYELOMA	PLASMA CELL LEUKEMIA
PLASMA CELLS IN BONE MARROW	<10%	>10%	>10%	>20% circulating plasma cells in Peripheral blood
M COMPONENT	<30 g/L	>30 g/L	Any level	High
EVIDENCE OF END ORGAN INVOLVEMENT	Nil	Nil	Any 1 CRAB/Slim	Present
MGUS: MONOCLONAL GAMMOPATHY OF UNKNOWN SIGNIFICANCE				

PROGNOSIS

- **Durie - Salmon** staging used earlier
- International staging system used now
 - Albumin and urine beta 2 microglobulin levels

T/t



00:38:33

WALDENSTROM MACROGLOBULINEMIA

- Excess production of **Ig M** → Severe Hyperviscosity manifestations
- No bony lytic lesions, no hypercalcemia or renal involvement
- Associated with **cold agglutinin syndrome** (e.g. of autoimmune hemolytic anaemia)
- **IOC** - bone marrow biopsy
- **T/t** - Ibrutinib (MOA: bruton tyrosine kinase inhibitor) + Rituximab

00:40:00



Important Information

- Δ are Ig G class, only variety with Ig M class - Cold agglutinin syndrome

HEAVY CHAIN DISEASES

HEAVY CHAIN DISEASES

Gamma chain overproduction: Franklin disease

Alpha chain overproduction: Seligman diseases (M/c)

POEMS SYNDROME

00:42:37

- Polyneuropathy
- Organomegaly
- Endocrinopathy (Hypothyroidism, Diabetes mellitus)
- M - M protein (monoclonal lambda chain of IgA or IgG)
- Skin changes - Hyperpigmentation, Acrocyanosis



Q. A 58-year-old male presents with fatigue, bone pain, and elevated serum protein levels. Bone marrow biopsy reveals 40% plasma cells. Cytogenetic studies show the following abnormalities: t(4:14), del 17p and gain of 1q. based on the Msmart 3.0 classification, which of the following best describes the patient's risk stratification?

- a. Standard risk myeloma
- b. Double hit myeloma
- c. High risk myeloma
- d. Triple hit myeloma

Ans. (d)

Q. A 58-year-old male presents with fatigue, bone pain, and recurrent infections. Lab test show elevated beta 2 microglobulin and low albumin levels. Which of the following drugs is not part of his initial therapy for transplant eligible patients?

- a. Bortezomib
- b. Lenalidomide
- c. Cyclophosphamide
- d. Dexamethasone

Ans. (c)



6. BLOOD COMPONENTS

BLOOD COMPONENTS

- 1 SDP (Single Donor Platelets) = 4-6 units PRP (Platelet Rich Plasma)
 - Incidence of allergic reactions ↑ with PRP as compared to SDP

COMPONENT	INDICATION OF TRANSFUSION	EFFICACY
RBC Concentrate	• Hb < 7 gm in hemodynamic stability to prevent chances of CHF • Hb < 8 gm in cardiac surgery, orthopedic surgery • Exchange transfusion: Sick cell crisis presents as acute chest syndrome • Thalassemia major	• Hb ↑: 1 gm% /unit • Hct ↑: 3%
Platelet Concentrate (PC) PC C/I in - • ITP • Thrombotic Microangiopathy • Heparin induced thrombocytopenia	• Platelet count < 5k/μl in absence of fever/infection • Platelet count < 10k/μl in presence of fever/infection • Platelet count < 50k/μl in surgery, DIC, endoscopy (invasive procedures) • Platelet count < 80,000/μl in neuro and eye surgeries	• Rise of platelets > 10 x 10⁹ platelets/L usually in 1st hour post transfusion

PLASMA

COMPONENT	INDICATION OF TRANSFUSION
Plasma	• Acute hypovolemic coagulopathy: DIC, Burns, fulminant hepatic failure
Plasma exchange (Plex)	• Guillain barre syndrome • Thrombotic thrombocytopenic purpura
Multicomponent PRBC, PC, Plasma (massive transfusion- whole blood volume replaced in 24 hours or 50% blood volume replaced in 3 hours)	• Acute hypovolemic coagulopathy ◦ Bleeding → hypothermia → less active clotting factors • RBCC: plasma: single donor platelets = 1:1:0.25 • 6 units of packed RBC, 6 units of plasma, 1 unit of single donor platelets



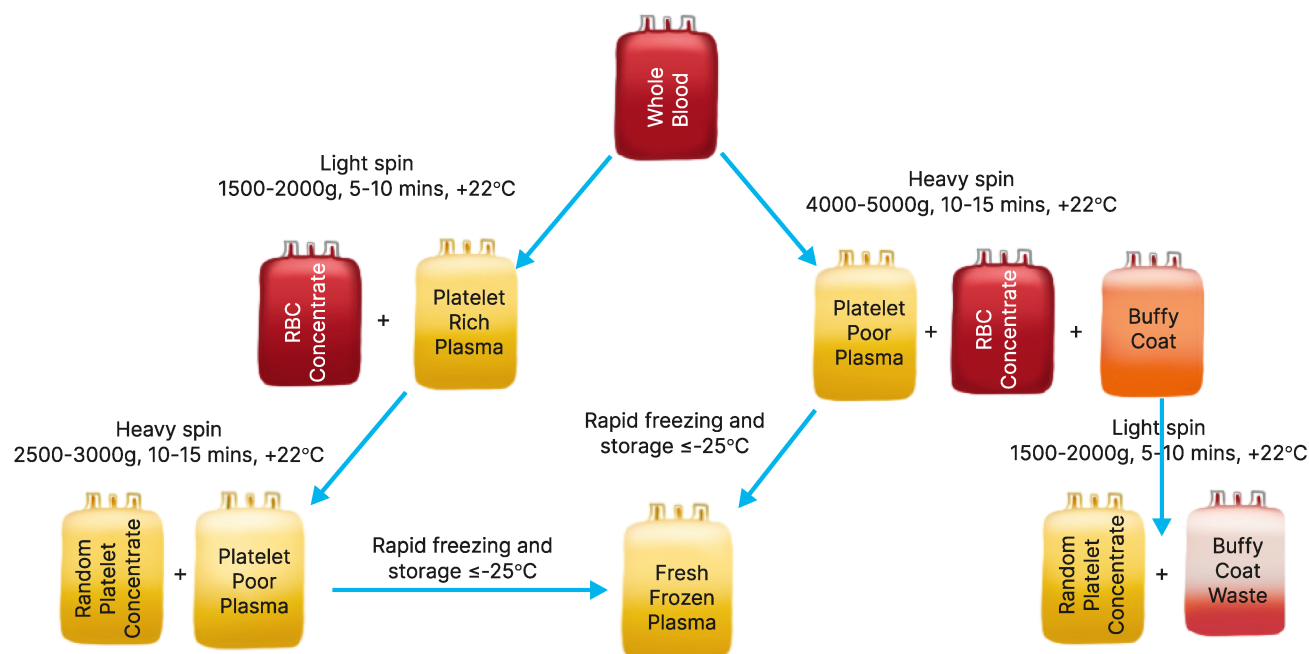
Granulocyte concentrate (doubts in efficacy)

- Severe bacteria/fungal infections
- Absolute neutrophil count $<100 \mu\text{l}$
- Dysfunctional granulocytes (CGD)

Cryoprecipitate (lacks factor IX)

- Risk of transmission of infection present

- Acute bleeding coagulopathy
 - Type II (Dysfunctional protein), III Von Willebrand Diseases (deficiency present)
 - Hemophilia A in absence of factor VIII concentrate



Algorithm for blood components preparation from whole blood

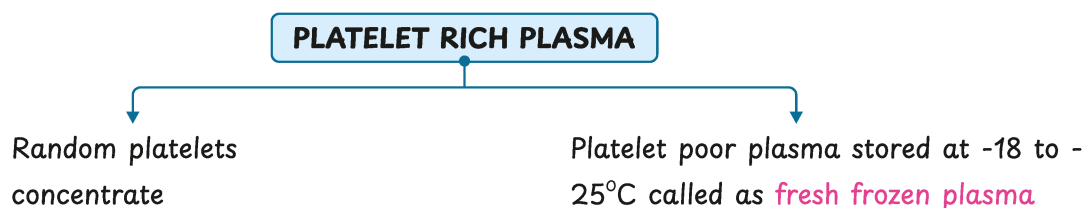
- Blood donation → centrifugation

Slow spin → generates
2 components

Fast spin → 3 components

- Slow spin/light spin-generates

- RBC concentrate → used in case of severe anaemia
- Platelet rich plasma

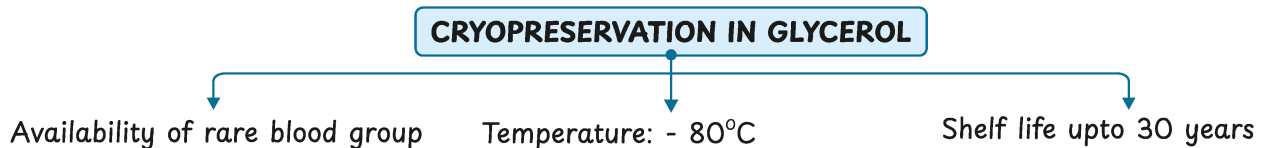


- Heavy spin done subsequently to segregate plasma from platelets
- Fractionation with heavy spin → 3 components produced
 - Platelet poor plasma
 - RBC concentrate
 - **Buffy coat**: Granulocytes + platelets → light spin done to segregate granulocytes and platelets

RBCC

00:11:38

- Made from whole blood or Apheresis and subjected to following processes
 - Deleucocytation: $<1-5 \times 10^6$ WBC/unit
 - Irradiation with X-ray/Gamma rays - **25 Gy** to reduce risk of Transfusion associated GVHD
 - Plasma reduction to reduce allergic reactions



- Volume of RBCC: **250-300 ml**
- Storage temperature and duration
 - Temperature: $4^{\circ}\text{C} \pm 2^{\circ}\text{C}$
 - Shelf life of RBC concentrate in blood bank: 42 days with SAGM and 35 days with CPDA (citrate phosphate dextrose agar)

PLATELET TRANSFUSION

00:13:48

- Pooled from 4 to 6 unit (PRP) of ABO identical units or apheresis (SDP)
 - Suspended in additive solution
 - Deleucocytation $<1-5 \times 10^6$ leukocytes/unit
 - Pathogen reduction
 - UV illumination
 - Nucleic acid cross linker
 - As platelets are stored at room temperature ($20-24^{\circ}\text{C}$), chances of bacterial transmission/transfusion transmitted infections is highest
- Volume
 - 100-700 ml
 - $>2 \times 10^{11}$ platelets/unit
- Storage
 - Temperature - $20-24^{\circ}\text{C}$ under permanent motion - 3-7 days (average 5 days)

PLASMA

00:15:15

- Storage temperature: **-18°C**
- Shelf life: **1-2 years**
- Volume: 200-300ml
- Contains coagulation factors
 - Fibrinogen
 - Anticoagulants- Protein C/S, AT III
- Used in: Fulminant hepatic failure, DIC, to manage acute hypovolemic coagulopathy

GRANULOCYTE CONCENTRATE

00:15:53

- Generated From whole blood/Apheresis
- Irradiation is mandatory: TA-GVHD↓ (can cause death in 3 weeks)
- Apheresis: 650ml, 2×10^{10} granulocyte/unit
- Storage: room temperature
- Used within < 24 hrs of **end** of collection
- Used in febrile neutropenia with ANC < 100 cells/cu mm

CRYOPRECIPITATE

00:17:17

- Collected after thawing of centrifugation of plasma
- Contain factor VIII, Fibrinogen, VWF (**lacks factor IX**)
- After thawing can be used within 6 hours
- Storage 12 months
- kept at room temperature for 6 hours

MCQ

- Q. 23-year-old man is undergoing induction chemotherapy for AML. He is having high grade fever and hypotension on day 10 of admission. Blood culture show Klebsiella and he is on Piperacillin and Tazobactam. ANC is 100 cells/cu.mm. bone marrow show regenerating myeloid precursors. Which of the following will be best for management of this patient?
- a. G-CSF (Filgrastim)
 - b. GM-CSF (Sargramostim)
 - c. Granulocyte concentrates
 - d. Add meropenem and colistin

Ans. **(c)**



COMPLICATIONS OF BLOOD TRANSFUSION

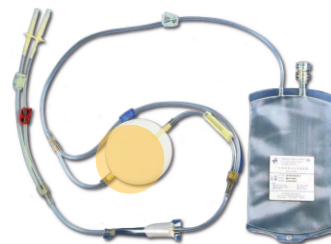
COMPLICATIONS OF BLOOD TRANSFUSION

STRATEGY TO REDUCE ADVERSE REACTIONS TO BLOOD COMPONENTS

00:00:33

- Rationale -
 - To generate Leucodepleted blood /Deleucocytation (99.99% of WBCs are prevented from entering body of recipient)
- Definition: < 1-5 X10⁶ donor leucocytes/unit
- Endpoints achieved -
 - Febrile Non-Hemolytic Transfusion Reaction ↓
 - CMV (Intracellular pathogenesis) ↓
 - Alloimmunization ↓
 - Immunomodulation ↓

LEUKOCYTE FILTER



- Rationale -
 - Collects initial 20 ml outflow of blood from donor and contains skin debris
- End point achieved -
 - ↓ contamination with skin flora
 - Transfusion transmitted bacterial infection

DIVERSION POUCH

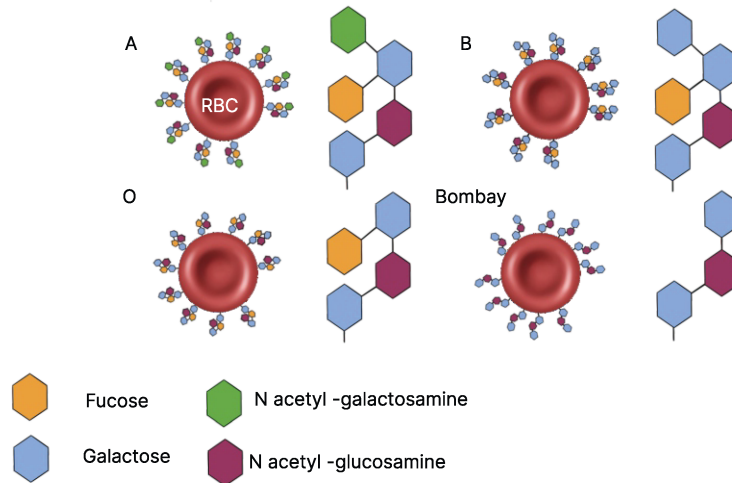


BLOOD GROUP ANTIGENS AND ANTIBODIES

00:04:56

- ABO gene on chromosome: 9, codominant pattern

BLOOD GROUPS	ANTIGENS	ENZYMES	IMMUNODOMINANT SUGARS
A	H (Fucose + glycolipid)	A transferase	N-acetyl galactosamine
B	H (Fucose + glycolipid)	B transferase	Galactose
AB	H	A and B transferase	N-acetyl galactosamine and galactose
O	H	Inactive enzymes	-



Important Information

- Rarest blood group (India): Bombay blood group, contains anti-A, anti-B, Anti H antibodies → on receiving blood from any other blood group → hemolytic reaction
- Universal plasma donor - AB
 - It lacks Anti A and Anti B antibodies

- Total 43 blood group systems
- Only 2 antigens are matched in blood grouping and cross matching out of 43 antigens → allergic reactions ↑

PROTEIN BLOOD GROUPS

- Rh system - D antigen (**Chromosome 1**)
 - D antigen present: Rh positive
 - D antigen absent: Rh negative
 - Husband is B + and wife is B -
→ Fetomaternal leak → production of antibodies → Erythroblastosis fetalis

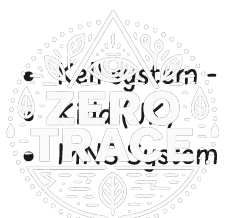
Antibodies produced in 1st pregnancy - **Ig M** class

During 2nd/3rd pregnancy → already sensitized immune system of mother → **Ig G** class → can cross placenta and reach baby → hemolysis → anemia and heart failure

→ Child is at risk of development of **immune hydrops fetalis** → can lead to development of **pathological jaundice**

- Duffy system
 - ↑ susceptibility to Plasmodium vivax
 - If absent → resistance to Plasmodium vivax
 - Nomenclature for Duffy system - **Fy antigen**

Fy antigen positive	↑ risk of development of plasmodium vivax infection
Fy antigen negative	Resistance to Plasmodium vivax



- Kall system - KEL 1
- Kall (K)
- InAS System

Q. In emergency situations, Can D + RBCs be transfused to D- individuals who do not have anti-D antibodies?

Ans. YES, in emergency can be given with proper documentation regarding consent taken for life saving step taken

ADDITIVE SOLUTIONS

00:14:41

	CONSTITUENTS	SHELF LIFE
SAGM	• Saline • Adenine: Maintain ATP component • Glucose ◦ Nutrition of RBC ◦ Maintain 2,3 DPG • Mannitol: Maintains RBC integrity	42 days
CPD-A	• Citrate: binds to calcium and help in preventing clotting of blood as calcium is co enzyme in clotting cycle • Phosphate: generation of 2,3 DPG • Dextrose • Adenine	35 days
CPD	Citrate, Phosphate, Dextrose	28 days
ACD	Acid, citrate, phosphate	21 days

Blood stored in blood bank at 4°C (2-6°C)

MASSIVE TRANSFUSION DEFINITION AND COMPLICATIONS

00:18:27

DEFINITION

- Replacement of >50% of total blood volume within **3 hours**
- >1 blood volume / 5L transfused in 24 hours
- >10 units of PRBC in 24 hours
- >4 units of PBC in 1 hour with ongoing need

RATIO OF BLOOD UNITS INFUSED

6 PBC (Packed RBC)	6FFP (Fresh frozen plasma)	6PRP (Platelet rich plasma)	1:1:1
PRBC	FFP	SDP	1:1:0.25

- 1 SDP (Single donor platelets) ~ **4-6 units of platelet** rich plasma